

Beth Israel Deaconess Medical Center

Sleep Disorders Center

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POLYSOMNOGRAPHY REPORT

Patient Name: Patient CB70E6AE
Date of Birth: October 8, 2086
Age: 61 years
Sex: Male
Medical Record Number: CB70E6AE-001
Study Date: April 12, 2148
Report Date: April 15, 2148
Referring Physician: Dr. Margaret L. Chen, MD – Internal Medicine
Sleep Physician: Dr. Jonathan R. Silverman, MD – Sleep Medicine
Sleep Technologist: Rachel M. Nguyen, RPSGT

Clinical History and Indication

This 61-year-old male was referred for polysomnographic evaluation due to loud snoring, witnessed apneas, excessive daytime sleepiness, and non-restorative sleep. The patient reports a history of waking gasping for air and frequent nocturnal awakenings. His bed partner confirms loud snoring with observed pauses in breathing. The patient endorses morning headaches and significant daytime fatigue affecting daily activities.

Past medical history is significant for type 2 diabetes mellitus on insulin, hypertension, hyperlipidemia, hypothyroidism, coronary artery disease with prior angioplasty, paroxysmal atrial fibrillation on anticoagulation, chronic lymphocytic leukemia, presence of cardiac pacemaker, history of bariatric surgery, obesity, anxiety disorder, major depressive disorder, peripheral neuropathy, and glaucoma. The patient has a history of TIA without residual deficits and personal history of testicular malignancy.

Current Medications: Metoprolol, levothyroxine, insulin (Glargine), aspirin, pantoprazole, divalproex, fluoxetine, trazodone, gabapentin, acetaminophen, docusate, senna, heparin, warfarin, and latanoprost ophthalmic solution.

Epworth Sleepiness Scale: 15/24 (indicates excessive daytime sleepiness)

Study Protocol

Type: Diagnostic in-laboratory attended polysomnography (PSG)

The patient underwent overnight polysomnography in our sleep laboratory. Monitoring included electroencephalography (EEG: F4-M1, C4-M1, O2-M1), bilateral electrooculography (EOG), submental and bilateral anterior tibialis electromyography (EMG), electrocardiography (ECG), nasal/oral airflow via pressure transducer and thermocouple, thoracic and abdominal respiratory inductance plethysmography, snoring microphone, body position sensor, and continuous pulse oximetry.

Sleep Architecture

Parameter	Value	Normal Range
Time in Bed (TIB)	468.5 minutes	—
Total Sleep Time (TST)	312.0 minutes	360–480 min
Sleep Latency	8.5 minutes	<20 min
REM Latency	156.0 minutes	60–120 min
Sleep Efficiency	66.6%	>85%
Wake After Sleep Onset (WASO)	148.0 minutes	<30 min
Number of Awakenings	47	<15
Arousal Index	42.3 per hour	<15 per hour

Sleep Stage Distribution

Stage	Minutes	% of TST	Normal % Range
N1 (Light Sleep)	124.5	39.9%	2–5%
N2 (Light Sleep)	140.5	45.0%	45–55%
N3 (Deep Sleep)	15.5	5.0%	13–23%
REM (Dream Sleep)	31.5	10.1%	20–25%

Respiratory Events

Summary of Events

Event Type	Total Count	Index (per hour of sleep)
Obstructive Apneas	156	30.0/hr
Mixed Apneas	8	1.5/hr
Central Apneas	4	0.8/hr
Hypopneas	45	8.7/hr
Apnea-Hypopnea Index (AHI)	213 events	41.0/hr
Respiratory Effort Related Arousals (RERAs)	6	1.2/hr
Respiratory Disturbance Index (RDI)	219 events	42.2/hr

AHI by Sleep Position

Position	Time (min)	AHI (per hour)
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Supine	187.5	58.4/hr
Non-Supine	124.5	16.9/hr

AHI by Sleep Stage

Sleep Stage	AHI (per hour)
NREM (N1, N2, N3)	39.2/hr
REM	51.4/hr

Oxygen Saturation Data

Baseline SpO₂:	96%
Mean SpO₂ During Sleep:	91%
Lowest SpO₂:	78%
% Time SpO₂ <90%:	18.4% of TST
Oxygen Desaturation Index (ODI ≥3%):	38.5 events per hour
Oxygen Desaturation Index (ODI ≥4%):	32.1 events per hour
Cardiac Events	

Continuous electrocardiographic monitoring revealed underlying sinus rhythm with occasional ventricular pacing consistent with the patient's known cardiac pacemaker. No significant arrhythmias were noted during the study. Heart rate ranged from 52 to 94 beats per minute during sleep.

Periodic Limb Movements

Total Periodic Limb Movements (PLMs):	28
Periodic Limb Movement Index (PLMI):	5.4 per hour
PLMs with Arousal:	4
PLM Arousal Index:	0.8 per hour
The periodic limb movement index is within normal limits (<15 per hour). Leg movements do not significantly contribute to sleep disruption.	

Snoring and Body Position

Loud snoring was documented throughout the study, particularly in the supine position. The patient spent approximately 60% of sleep time in the supine position. Snoring was associated with respiratory events.

Interpretation

SEVERE OBSTRUCTIVE SLEEP APNEA SYNDROME

This diagnostic polysomnogram demonstrates severe obstructive sleep apnea with an overall Apnea-Hypopnea Index (AHI) of 41.0 events per hour (normal <5/hr). The sleep-disordered breathing is predominantly obstructive in nature, characterized by repetitive episodes of upper airway collapse during sleep.

Key findings include:

1. Severe Sleep-Disordered Breathing: The AHI of 41.0/hr indicates severe obstructive sleep apnea. The respiratory events are significantly worse in the supine position (AHI 58.4/hr) and during REM sleep (AHI 51.4/hr), consistent with positional and REM-related OSA.

2. Significant Oxygen Desaturation: The patient experienced clinically significant oxygen desaturation with a nadir of 78% and spent 18.4% of total sleep time with oxygen saturation below 90%. The oxygen desaturation index of 38.5/hr correlates closely with the frequency of respiratory events. These desaturations place the patient at increased risk for cardiovascular complications, particularly given his extensive cardiac history.

3. Severely Fragmented Sleep: Sleep efficiency is markedly reduced at 66.6% (normal >85%) with elevated wake after sleep onset (148 minutes) and 47 awakenings. The arousal index of 42.3/hr is significantly elevated, indicating profound sleep fragmentation predominantly related to respiratory events.

4. Abnormal Sleep Architecture: Sleep architecture is disrupted with marked increase in N1 sleep (39.9% vs. normal 2–5%), reduced slow-wave sleep (N3: 5.0% vs. normal 13–23%), and suppressed REM sleep (10.1% vs. normal 20–25%). The prolonged REM latency (156 minutes) and decreased REM percentage reflect the disrupting effect of sleep apnea on sleep stage progression.

5. Positional Dependency: The significantly higher AHI in the supine position (58.4/hr) compared to non-supine position (16.9/hr) indicates a strong positional component to this patient's sleep apnea.

The severe obstructive sleep apnea documented in this study likely contributes significantly to the patient's symptoms of excessive daytime sleepiness, non-restorative sleep, and morning headaches. Given the patient's extensive cardiovascular comorbidities (coronary artery disease, atrial fibrillation, hypertension, pacemaker), the repetitive hypoxemia and sympathetic activation from severe OSA represent substantial cardiovascular risk.

Recommendations

1. Positive Airway Pressure (PAP) Therapy: The patient requires treatment with continuous positive airway pressure (CPAP) or auto-adjusting positive airway pressure (APAP) therapy. A PAP titration study is strongly recommended to determine the optimal therapeutic pressure for elimination of obstructive events, snoring, and oxygen desaturation.

2. PAP Titration Study: Schedule in-laboratory CPAP titration polysomnography within the next 4–6 weeks to determine appropriate therapeutic pressure settings.

3. Sleep Hygiene and Weight Management: Continue weight management efforts through diet and exercise, as weight loss can improve OSA severity in the context of the patient's obesity and prior bariatric surgery history.

4. Positional Therapy Consideration: Given the marked positional component, adjunctive positional therapy to avoid supine sleeping may provide additional benefit in conjunction with PAP therapy.

5. Cardiovascular Risk Management: Results of this study have been communicated to the patient's cardiologist given the implications for cardiovascular risk management. Treatment of OSA with PAP therapy may improve blood pressure control, reduce atrial fibrillation burden, and decrease overall cardiovascular morbidity.

6. Follow-up: The patient should return to the Sleep Disorders Center for CPAP titration study and subsequent follow-up appointment 4–6 weeks after CPAP initiation to assess therapy adherence, efficacy, and symptomatic improvement.

7. Coordinate Care: Close coordination with cardiology, endocrinology, and primary care is recommended given the patient's complex medical history and multiple comorbidities that may be affected by untreated sleep apnea.

Diagnoses

1. Obstructive Sleep Apnea Syndrome, Severe (ICD-10: G47.33)

Apnea-Hypopnea Index: 41.0 events per hour

2. Sleep-Related Hypoxemia (ICD-10: G47.36)

Lowest oxygen saturation: 78%

3. Insomnia Related to Sleep-Disordered Breathing (ICD-10: G47.00)

Sleep efficiency: 66.6%

Report electronically reviewed and approved by:

Jonathan R. Silverman, MD, FAASM

Board Certified in Sleep Medicine and Internal Medicine

Medical Director, BIDMC Sleep Disorders Center

Date of Report: April 15, 2148

Time: 14:35 EST

Electronic Signature on File

Distribution: Patient Chart, Dr. Margaret L. Chen (Referring Physician), Patient Portal

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